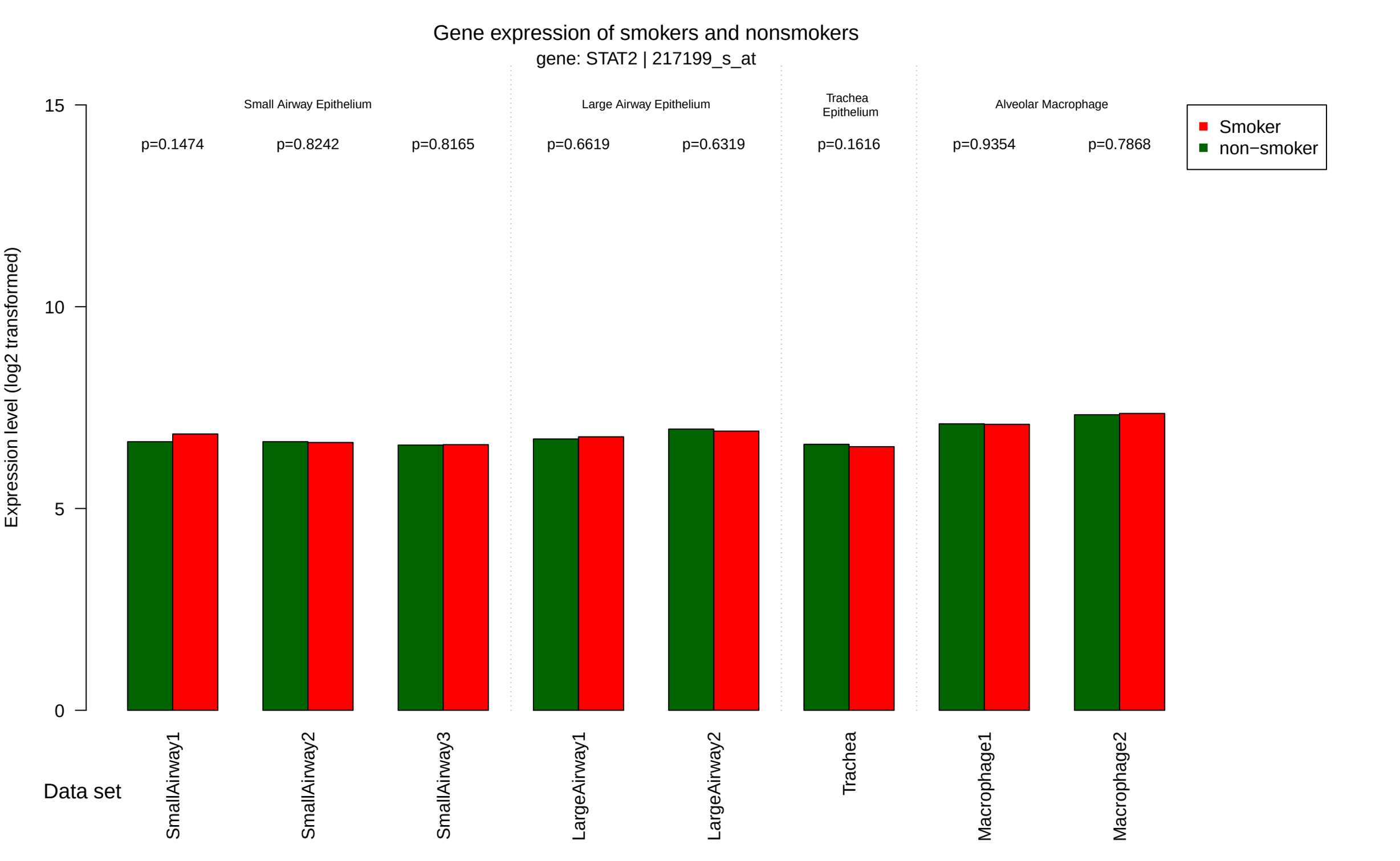
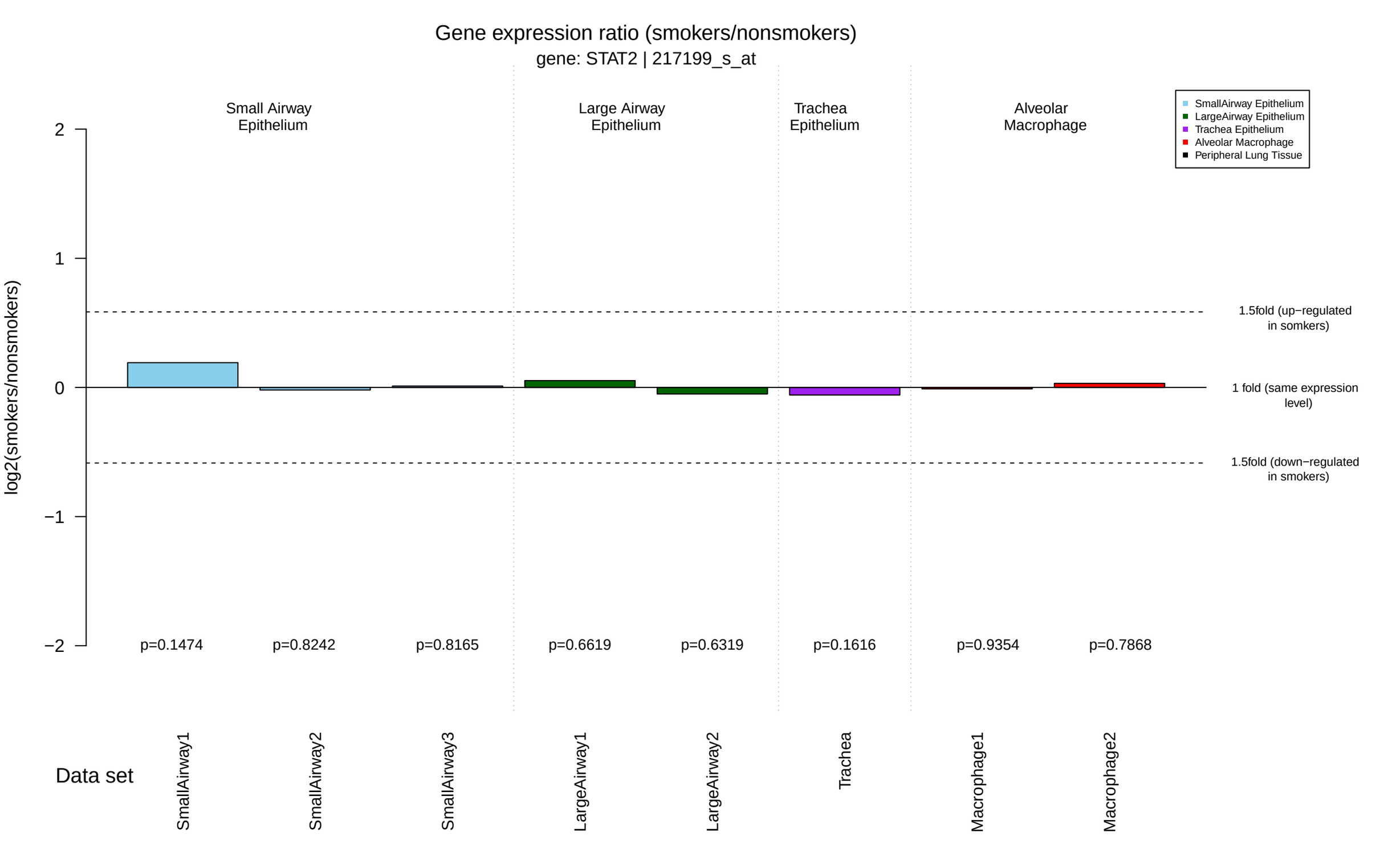
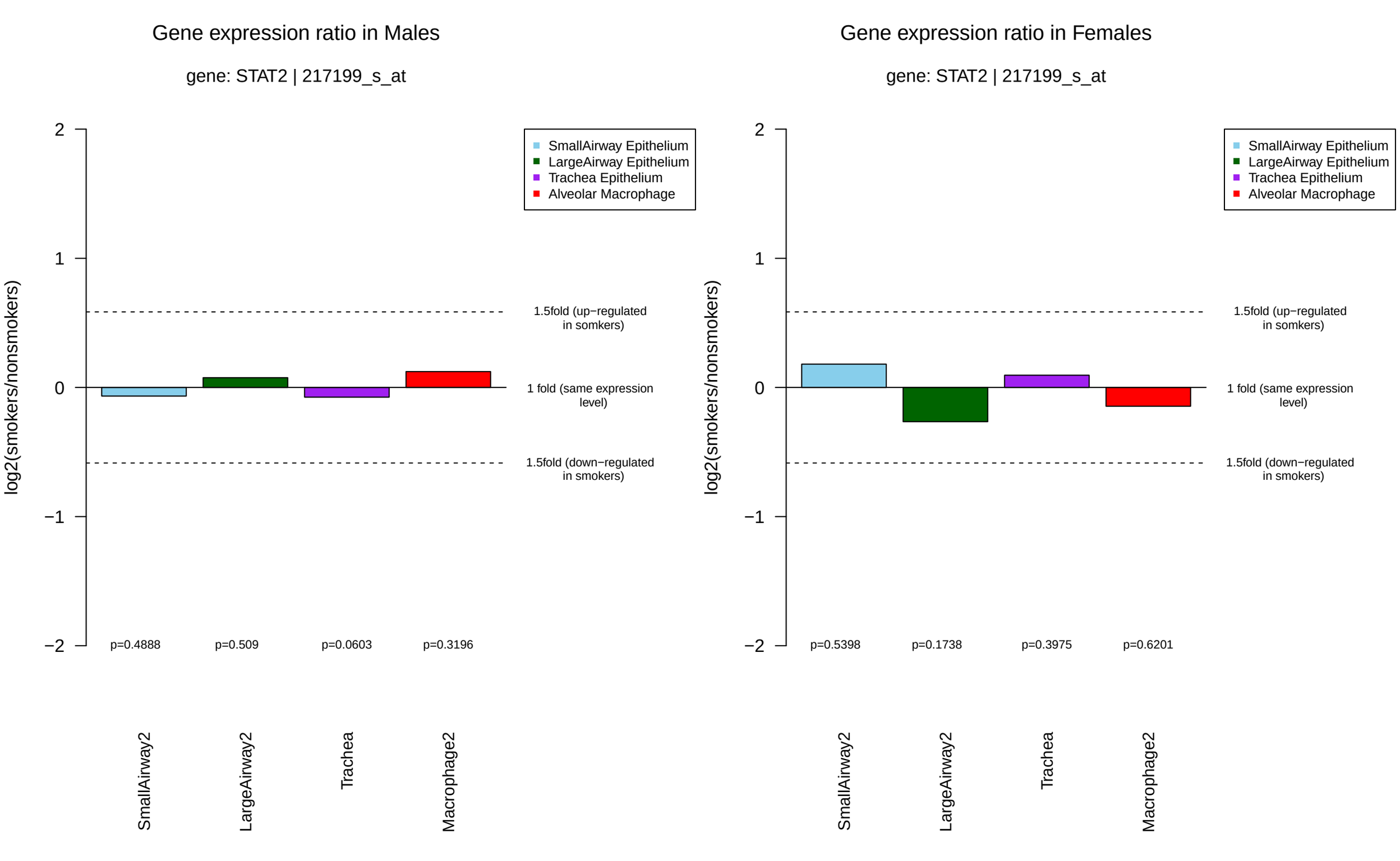
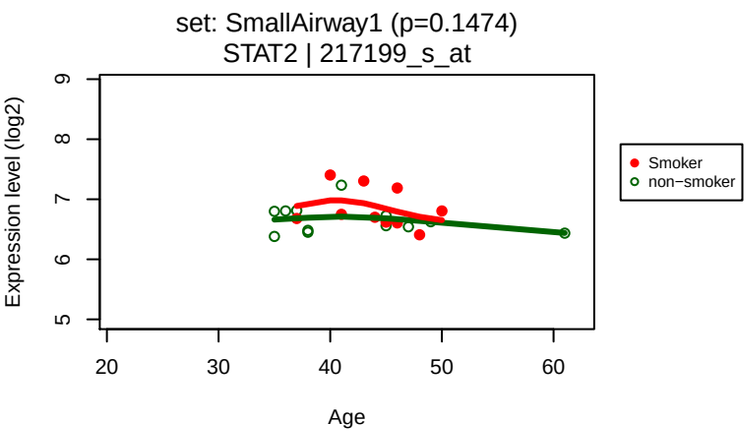
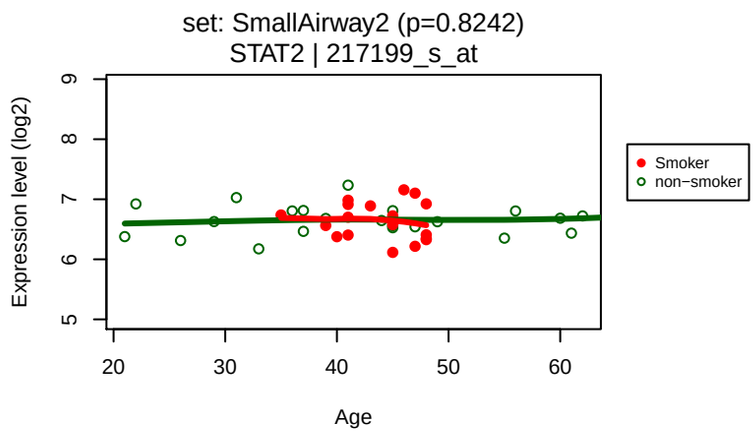
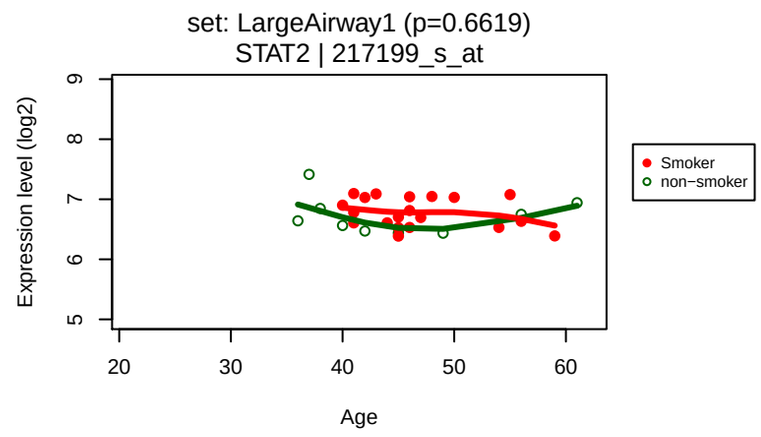

Gene expression of smokers and nonsmokers
gene: STAT2 | 217199_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1474
p=0.8242
p=0.8165
p=0.6619
p=0.6319
p=0.1616
p=0.9354
p=0.7868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

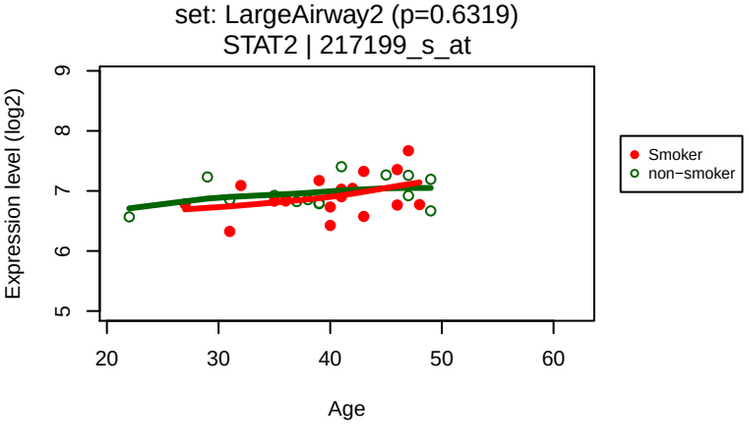

Gene expression ratio (smokers/nonsmokers)
gene: STAT2 | 217199_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1474
p=0.8242
p=0.8165
p=0.6619
p=0.6319
p=0.1616
p=0.9354
p=0.7868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

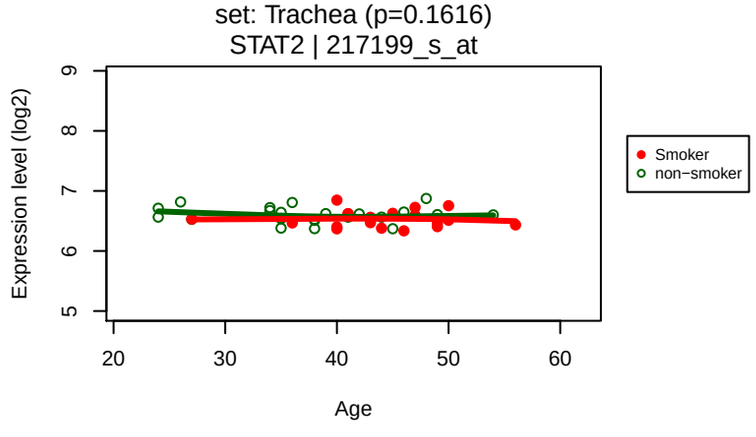

Gene expression ratio in Males
gene: STAT2 | 217199_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4888
p=0.509
p=0.0603
p=0.3196
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT2 | 217199_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5398
p=0.1738
p=0.3975
p=0.6201
SmallAirway2
LargeAirway2
Trachea
Macrophage2

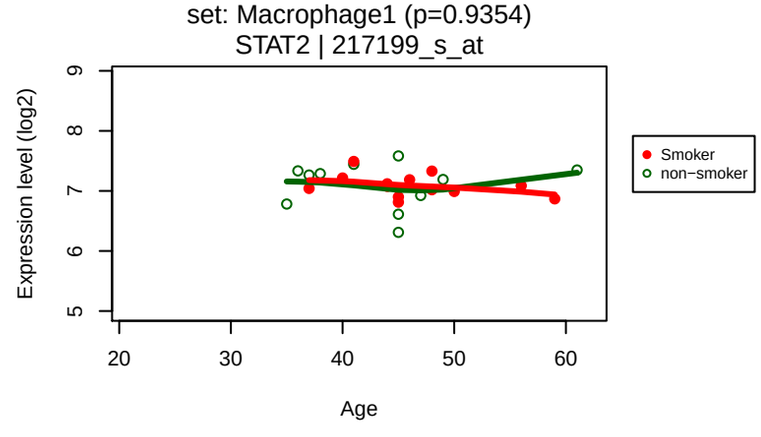

set: SmallAirway1 (p=0.1474)
STAT2 | 217199_s_at
Expression level (log2)
Age
Smoker
non−smoker

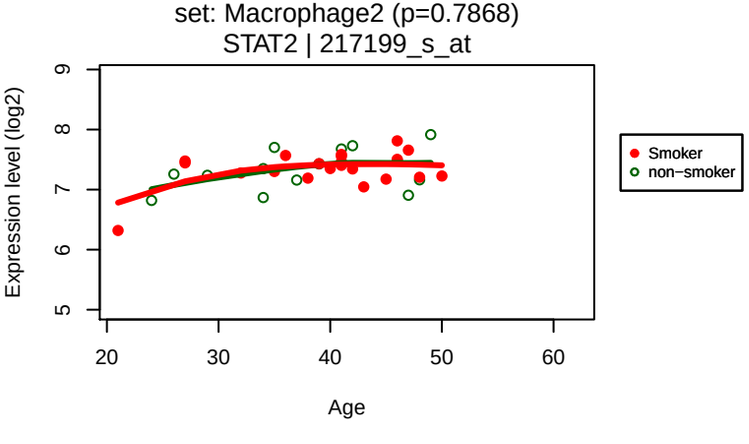

set: SmallAirway2 (p=0.8242)
STAT2 | 217199_s_at
Expression level (log2)
Age
Smoker
non−smoker

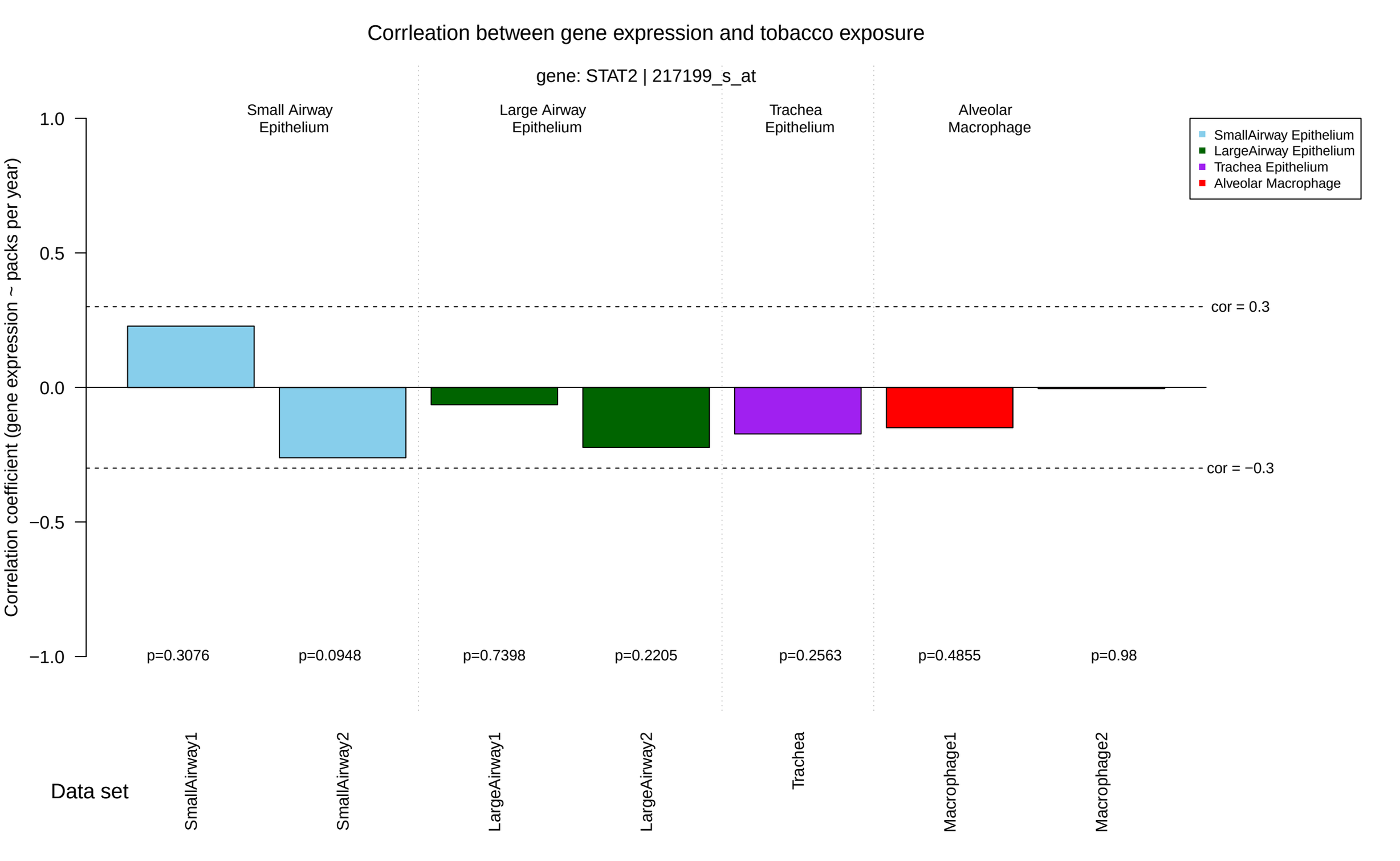

set: LargeAirway1 (p=0.6619)
STAT2 | 217199_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6319)
STAT2 | 217199_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1616)
STAT2 | 217199_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9354)
STAT2 | 217199_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7868)
STAT2 | 217199_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT2 | 217199_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3076
p=0.0948
p=0.7398
p=0.2205
p=0.2563
p=0.4855
p=0.98
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2